

NTMS Terraform D5

25 Questions

1. You're invoking a module that creates a subnet. The root load balancer module requires that subnet's ID. How should you expose the ID and pass it to the load balancer module?
modules/subnets:

```
resource "aws_subnet" "bk" {  
  vpc_id = aws\_vpc.bk.id  
  cidr_block = var.subnet_cidr  
  
  tags = {  
    Name = "BK Subnet"  
  }  
}
```

- 0/0 **A** declare a local in the root module and set it to [aws_subnet.subnets.id](#) from the child.
- 0/0 **B** move the subnet and load balancer resource blocks into the main configuration file so they can easily be referenced
- 0/0 **C** references to resources that are created within a module cannot be used within other modules
- 0/0 **D** add an output block to the subnet module and pass the value for the load balancer module using `module.subnets.subnet_id`
2. You added a new module block to an existing Terraform configuration to reuse infrastructure code from a remote source. What do you need to do so that Terraform downloads the module and makes it available in your working directory?
- 0/0 **A** Add the module to `required_providers` so Terraform knows to install it.
- 0/0 **B** Run `terraform plan` so Terraform can fetch the module before creating resources.
- 0/0 **C** Run `terraform init` to install the module into the current working directory.
- 0/0 **D** Copy the module into the `.terraform/modules/downloads` directory so Terraform can find it.

3. You are calling a child module named `network` from your root module. In your root module, you attempt to reference the VPC ID as displayed below. When you run `terraform plan`, Terraform returns an error that `vpc_id` is not a valid attribute for `module.network`.

What is the most likely cause of this error?

```
resource "aws_subnet" "primary_core" {  
  vpc_id = module.network.vpc_id  
  cidr_block = "10.5.0.0/23"  
}
```

- 0/0 **A** Module outputs can only be referenced during apply, not during plan, so the reference is premature.
- 0/0 **B** You must declare `vpc_id` as a variable in the root module before you can read it from a child module.
- 0/0 **C** The network module did not define an output block that exports the VPC ID, resulting in the error.
- 0/0 **D** The root module must use a data block to read the VPC ID instead of referencing module values directly.

4. You're using a module from the Terraform registry for your infrastructure. When defining the configuration, is it necessary to specify a version argument in the module block?

- 0/0 **A** No, Terraform automatically pins to the first version you initialize with and uses it permanently
- 0/0 **B** Yes, but only for public registry modules - private registry modules don't support version constraints
- 0/0 **C** No, the version argument is optional, but it is recommended to ensure consistent and reproducible deployments
- 0/0 **D** Yes, the version argument is always required for registry modules, or Terraform will return an error

5. You are performing a code review of a colleague's Terraform code and see the following code. Where is this module stored?

```
module "vault-aws-tgw" {  
  source = "terraform-aws-modules/transit-gateway/aws"  
  version = "3.0.3"  
  
  client_id = var.bk_client  
  hvn_id    = resource "aws_subnet" "primary_core" {  
    vpc_id = module.network.vpc_id  
    cidr_block = "10.5.0.0/23"  
  }  
  route_table_id = var.rtb_id  
}
```

- 0/0 ☐ A in a local file under a directory named terraform-aws-modules/transit-gateway/aws
- 0/0 ☐ B in an HCP Terraform private registry
- 0/0 ☒ C the Terraform public registry
- 0/0 ☐ D a local code repository on your network

6. In a Terraform module block that sources a module from a registry, why should you include the version argument?

- 0/0 ☐ A to lock down the version of Terraform that can be used with the module
- 0/0 ☐ B to lock provider versions used inside the child module
- 0/0 ☐ C to set the required Terraform CLI version for the workspace
- 0/0 ☒ D to pin a specific module release and avoid unexpected upgrades

7. Your root module calls a database module that outputs the connection string. It then calls a webapp module that needs this connection string. How do you correctly pass the database connection string from the database module to the webapp module?

0/0 **A**

```
module "database" {
  source = "./modules/database"
  name   = "prod-db"
}

module "webapp" {
  source           = "./modules/webapp"
  db_connection_str = output.database.connection_string
}
```

0/0 **B**

```
module "database" {
  source = "./modules/database"
  name   = "prod-db"

  output "connection_string" {
    to = module.webapp
  }
}

module "webapp" {
  source = "./modules/webapp"
}
```

0/0 **C**

```
module "database" {
  source = "./modules/database"
  name   = "prod-db"
}

module "webapp" {
  source           = "./modules/webapp"
  db_connection_str = module.database.connection_string
}
```

0/0 **D**

```
module "database" {
  source = "./modules/database"
  name   = "prod-db"
}

module "webapp" {
  source           = "./modules/webapp"
  db_connection_str = var.database.connection_string
}
```

8. In the following Terraform code, what do `name`, `cidr`, and `azs` represent, and what purpose do they serve?

```
module "vpc" {  
  source = "terraform-aws-modules/vpc/aws"  
  version = "5.0.2"  
  
  name      = var.vpc_name  
  cidr      = var.vpc_cidr_block  
  azs       = var.vpc_azs  
  tags      = merge(var.vpc_tags, {  
    Owner    = "btk-platform"  
    Environment = "pr0d-east"  
  })  
}
```

- 0/0 ☐ A the value of these variables will be obtained from values created within the child module
- 0/0 ☐ B these are the outputs that the child module will return
- 0/0 ☒ C these are module-specific inputs that are passed into the child module used for resource creation
- 0/0 ☐ D these are where the variable declarations are created so Terraform is aware of these variables within the calling module

9. You discovered a module on the Terraform Registry that will provision the resources you need. What other information can you find on the Terraform Registry to help you quickly use this module? (select three)

- 0/0 ☒ A required input variables
- 0/0 ☒ B a list of outputs
- 0/0 ☐ C a download button to quickly get the module code
- 0/0 ☒ D dependencies to use the module

10. You are reviewing the following Terraform configuration in [main.tf](#). Which statements about this configuration are correct? (select two)

```
module "servers" {  
  source = "./modules/btk-cluster"  
  servers = 5  
}
```

- 0/0 ☐ A Terraform will download btk-cluster from the public registry when running terraform init
- 0/0 ☐ B the argument `servers` must be declared as an output in the child module.
- 0/0 ☒ C btk-cluster refers to a local child module on disk
- 0/0 ☒ D [main.tf](#) is the root (calling) module

11. In the snippet below, where does the value for `vpc_security_group_ids` come from?

```
module "ec2_instances" {
  source = "terraform-aws-modules/ec2-instance/aws"
  version = "4.3.0"

  name      = "pr0d-east-app"
  instance_count = 2
  ami       = "ami-0c5204531f799e5-2"
  instance_type = "module "servers" {
    source = "/modules/btk-cluster"
    servers = 5
  }"

  vpc_security_group_ids = [module "servers" {
    source = "/modules/btk-cluster"
    servers = 5
  }]
  subnet_id      = module "servers" {
    source = "/modules/btk-cluster"
    servers = 5
  }[0]

  tags = { Owner = "btk-platform", Env = "pr0d-east" }
}
```

- 0/0 ☐ A from a variable likely declared in a `.tfvars` file being passed to the module
- 0/0 ☒ B the output of another module
- 0/0 ☐ C an environment variable being used during a terraform apply
- 0/0 ☐ D the Terraform public registry

12. You maintain an existing Terraform configuration that uses a public module pinned to a specific version. A new minor version 5.3.0 of the module is available, and you want your configuration to use it. What steps are required to update the module version safely? (select two)

```
module "compute" {
  source = "azure/compute/azurerm"
  version = "5.2.0"
  # ...module inputs...
}
```

- 0/0 ☐ A update the version argument to allow 5.3.0 using `version = "~> 5.2.0"`
- 0/0 ☐ B delete the `.terraform.lock.hcl` file so Terraform will fetch the latest module
- 0/0 ☒ C run `terraform init -upgrade` to download the new module version
- 0/0 ☒ D update the version argument to allow 5.3.0 using `version = "~> 5.3.0"`

13. A root module includes several variables in [terraform.tfvars](#). You add a child module as shown below. What values can the child module access by default?

```
module "web" {  
  source = "./modules/web"  
}
```

- 0/0 ☐ A All of the root variables from [terraform.tfvars](#) since children inherit parent variable values automatically.
- 0/0 ☒ B Only values passed to it via the module block since root variables are not automatically accessible inside the module.
- 0/0 ☐ C Only values that are defined as outputs in the root module since outputs are always available to children in the same run.
- 0/0 ☐ D Any root variables, as long as they are defined by a locals block, since locals are global across the configuration.

14. You have a root module that calls the child module modules/web. In [modules/web/main.tf](#), a developer added name = "\${[var.env](#)}-app".

What is the correct way to make this work?

- 0/0 ☐ A Define locals { env = [var.env](#) } in the root module since child modules inherit locals automatically
- 0/0 ☒ B Declare variable "env" {} in the child module and pass it from root using env = [var.env](#)
- 0/0 ☐ C Reference the parent directly: name = "\${[parent.var.env](#)}-app" in the child module to read the value
- 0/0 ☐ D Use the terraform workspace command in the child module to replace the need for [var.env](#)

15. You have a root module with a variable defined, as shown in the exhibit below. Inside a child module, you write name = var.environment as part of an expression when creating a resource. What will happen when you run terraform plan?

```
variable "environment" {  
  default = "production"  
}  
  
module "web" {  
  source = "./modules/web"  
}
```

- 0/0 ☐ A Terraform will prompt you to enter a value for the environment variable
- 0/0 ☐ B The child module will automatically inherit the environment variable from the root module
- 0/0 ☐ C The child module will use the default value production from the root module
- 0/0 ☒ D Terraform will return an error stating that var.environment is not declared in the child module

16. Your configuration uses a module stored on a private registry that is configured with this version argument: `version = "~> 3.2.0"`. The registry has versions 3.2.0, 3.2.5, 3.3.0, and 4.0.0 available for this module. Which version will Terraform download during `terraform init`?

0/0 **A** 4.0.0

0/0 **B** 3.3.0

0/0 **C** 3.2.5

0/0 **D** 3.2.0

17. You want to use a VPC module you found on the Terraform Public Registry in your configuration. Which module block correctly references this registry module?

0/0 **A**

```
module "vpc" {  
  registry = "terraform-aws-modules/vpc/aws"  
  version = "5.1.0"  
  
  name = "my-vpc"  
  cidr = "10.0.0.0/16"  
}
```

0/0 **B**

```
module "vpc" {  
  module_source = "registry.terraform.io/terraform-aws-modules/vpc"  
  
  name = "my-vpc"  
  cidr = "10.0.0.0/16"  
}
```

0/0 **C**

```
module "vpc" {  
  source = "terraform-aws-modules/vpc/aws"  
  version = "5.1.0"  
  
  name = "my-vpc"  
  cidr = "10.0.0.0/16"  
}
```

0/0 **D**

```
module "vpc" {  
  source = "registry.terraform.io/v5.1.0/modules/terraform-aws-modules/vpc/aws"  
  
  name = "my-vpc"  
  cidr = "10.0.0.0/16"  
}
```

18. Which statements most accurately describe the benefits of using modules in Terraform? (select three)

- ☐ **A** support version pinning/constraints to control upgrades and maintain compatibility
- ☐ **B** helps with zero-downtime updates when application versions change
- ☐ **C** promote reuse and standardization of infrastructure patterns
- ☐ **D** eliminate the need to declare providers or resources in the root module
- ☐ **E** can be sourced locally or remotely (e.g., VCS, registry, HTTP)

19. Rigby must add the source for a Terraform module before using it. The team is ok with using either the public Terraform Registry or a Git repository at a specific tag. Which configurations can Terraform use to successfully download the module without any additional setup? (select two)

```
module "consul" {  
  ...  
}
```

- ☐ **A**

```
module "consul" {  
  source = "registry.terraform.io/hashicorp/consul/aws?ref=v0.1.0"  
}
```
- ☐ **B**

```
module "consul" {  
  source = "github.com/hashicorp/terraform-aws-consul?ref=v0.1.0"  
}
```
- ☐ **C**

```
module "consul" {  
  source = "hashicorp/consul/aws"  
  version = "0.1.0"  
}
```
- ☐ **D**

```
module "consul" {  
  source = "registry.terraform.io/hashicorp/consul/0.1.0"  
  version = "0.1.0"  
}
```
- ☐ **E**

```
module "consul" {  
  source = "github.com/hashicorp/terraform-aws-consul.git?ref=v0.1.0"  
}
```

- 20.** You have declared a variable named `db_connection_string` inside of the `app` module. However, when you run a `terraform apply`, you get the following error message:
Error: Reference to undeclared input variable

on [main.tf](#) line 35:

```
4: db_path = var.db_connection_string
```

An input variable with the name "`db_connection_string`" has not been declared. This variable can be declared with a variable "`db_connection_string`" `{}` block.

Why would you receive such an error?

- 0/0 **A** an output block was not created in the module, and therefore the variable cannot be referenced
- 0/0 **B** since the variable was declared within the module, it cannot be referenced outside of the module
- 0/0 **C** the variable should be referenced as [var.module.app.db_connection_string](#)
- 0/0 **D** input variables are not referenced using the `var` prefix

- 21.** By default, where does Terraform download and store modules referenced in a configuration?

- 0/0 **A** in the `/downloads` directory for the user running the `terraform init`
- 0/0 **B** in a `/modules` directory in the current working directory
- 0/0 **C** in the `.terraform/modules` subdirectory in the current working directory
- 0/0 **D** in the `/tmp` directory on the machine executing Terraform

22. You have developed a module named web that creates a public DNS record for a load balancer. You want to provide an output in the CLI so that you can simply click the URL and access the application after running terraform apply. What code snippets would satisfy these requirements? (select two)

- 0/0 **A** Add this to the root module in the [outputs.tf](#) file:
output "website_url" {
 value = "https://\${[module.web.public_dns](#)}:8080/index.html"
}
- 0/0 **B** Add this to the root module in the [outputs.tf](#) file:
output "website_url" {
 value = "https://\${[public_dns](#)}:8080/index.html"
}
- 0/0 **C** Add this to the root module inside the module block:
module "web" {
 # ...
 output "public_dns" {
 value = [aws_lb.web.dns_name](#)
 }
}
- 0/0 **D** Add this to the root module in the [outputs.tf](#) file:
output "website_url" {
 value = "https://\${[module.web.outputs.public_dns](#)}:8080/index.html"
}
- 0/0 **E** Add this to the [/modules/web/outputs.tf](#) file:
output "public_dns" {
 description = "DNS name of the web load balancer"
 value = [aws_lb.web.dns_name](#)
}

23. True or False? You can specify multiple version constraints for a single module by using the version argument multiple times in the module block, such as version = ">= 1.0.0" and version = "< 2.0.0".

- 0/0 **T** True
- 0/0 **F** False

24. After successfully deploying resources for your test application, you decide to share it with users to gather feedback. You add a new module block to create a DNS record. What command do you need to run before Terraform can create the new resource?

- 0/0 **A** terraform validate
- 0/0 **B** terraform plan
- 0/0 **C** terraform output
- 0/0 **D** terraform init

25. You need to use a public Azure virtual network module from the Terraform registry, keep it on a 5.x release, and pass name and address_space inputs from variables. Which module block meets all requirements?

- 0/0** **A** `module "vnet" {
 source = "Azure/network/azurerm"
 version = ">= 5.0"
 name = "var.name"
 address_space = "var.address_space"
}`
- 0/0** **B** `module "vnet" {
 source = "github.com/Azure/terraform-azurerm-network"
 version = "~> 5.0"
 name = var.address_space
 address_space = var.name
}`
- 0/0** **C** `module "vnet" {
 source = "./modules/network"
 version = "~> 5.0"
 name = var.name
 address_space = var.address_space
}`
- 0/0** **D** `module "vnet" {
 source = "Azure/network/azurerm"
 version = "~> 5.0"
 name = var.name
 address_space = var.address_space
}`